

BANARAS HINDU UNIVERSITY

B.Sc. (Hons.) Semester IV Examination, 2013-14 (Old Course)

Paper No. BPT-401: Electromagnetic Theory and Basic Electronics

Time: 3 Hours

Maximum Marks: 70

*Instructions: Answer **five** questions including Question 1 which is compulsory. Symbols have their usual meanings.*

Question 1. Answer any **five** of the following: [5 × 4 = 20]

- (a) Prove that the divergence of the curl of any vector field vanishes: $\nabla \cdot (\nabla \times \vec{A}) = 0$.
- (b) Define scalar and vector potentials. How do you express the electric field $\vec{E}$ and magnetic field $\vec{B}$ in terms of these potentials?
- (c) Write down the four Maxwell's equations in differential form and briefly explain the physical meaning of each.
- (d) A material has conductivity $\sigma = 10^{-3} \text{ S/m}$ and relative permittivity $\epsilon_r = 2$. At what frequency would the conduction current density be equal to the displacement current density?
- (e) Describe the position of the Fermi level in an intrinsic and an extrinsic semiconductor. Discuss the effect of temperature on its position.
- (f) Why is a capacitor filter mostly used in a power supply rather than an inductor filter? Explain.
- (g) Explain how the current gain α is different from β . Establish the relationship between α and β for a transistor.

Question 2.

- (a) State and prove Gauss's divergence theorem. [5]
- (b) Derive the electromagnetic wave equation for the propagation of a plane wave in a non-conducting isotropic dielectric medium. Obtain expressions for the propagation constant, intrinsic impedance, phase velocity, and group velocity. [9]

Question 3.

- (a) State and prove Poynting's theorem for the conservation of energy in an electromagnetic field, and discuss the physical significance of each term in the equation. [8]
- (b) Define depth of penetration (skin depth). Show that electromagnetic waves are heavily attenuated inside a good conductor. [6]

OR

- (a) A plane wave travelling in a homogenous, isotropic, and linear dielectric medium is incident obliquely at a plane interface of another dielectric. Determine the reflection and transmission coefficients if the incident wave is polarized with its electric vector lying perpendicular to the plane of incidence. [9]
- (b) What is a transmission line? Explain the different parameters of a transmission line. Determine the characteristic impedance of a lossless cable if the inductance per unit length $L = 0.25 \mu\text{H/m}$ and capacitance per unit length $C = 100 \text{ pF/m}$. [5]

Question 4.

- (a) Explain the depletion region in a p-n junction. Show that the width of the depletion region in the p-n junction decreases with the increase of doping impurity concentration. [8]
- (b) Describe the working of a single-stage RC coupled common-emitter amplifier. Plot the frequency response of the amplifier and explain its features. [6]

OR

- (a) Explain the working of a full-wave bridge rectifier having a capacitor filter with a suitable circuit diagram. Obtain the relationship for the ripple factor. [8]
- (b) Why does a transistor need biasing? Draw the circuit diagram of a transistor self-bias (voltage divider bias) circuit. Explain how it provides better thermal stability than a fixed-bias circuit. [6]

Question 5.

- (a) Starting from Maxwell's equations, derive the wave equation for the propagation of a plane electromagnetic wave in free space. Obtain the solution representing a plane wave propagating along the $+z$ direction. [7]
- (b) Find the Brewster angle for a parallel-polarized wave travelling from air into glass for which $\epsilon_r = 5.0$. [4]
- (c) Explain the meaning of the Early effect. How does it affect the collector current in a junction transistor? [3]